

1. Explain OSI Reference Model

Answer:

OSI stands for Open Systems Interconnection model. It is a 7-layer reference model developed by ISO to standardize network communication.

The 7 layers are:

Layer	Function	Examples
Application	User services	HTTP, FTP
Presentation	Encryption, compression	SSL
Session	Establish session	NetBIOS
Transport	Reliable delivery	TCP, UDP
Network	Routing	IP
Data Link	Framing, MAC addressing	Ethernet
Physical	Transmission of bits	Cables

Data flows from Application layer to Physical layer at sender side and reverse at receiver side.

2. Size of IPv4 and MAC Address

Answer:

Address	Size
IPv4	32 bits
MAC Address	48 bits

IPv4 example:

192.168.1.1

MAC example:

00:1A:2B:3C:4D:5E

3. Explain Layerwise Protocols

Answer:

OSI Layer	Protocols
Application	HTTP, FTP, SMTP, DNS
Transport	TCP, UDP
Network	IP, ICMP
Data Link	Ethernet
Physical	Fiber optic, cables

4. What do you mean by Socket? Explain Socket Header Files

Answer:

Socket is an endpoint for communication between two systems over a network.

Socket programming is used for client-server communication.

Important header files:

Header File Purpose

sys/socket.h Socket functions

netinet/in.h Internet address structures

arpa/inet.h IP conversion

unistd.h Close function

5. Explain Functions used in Socket Programming**Answer:****Function Purpose**

socket() Create socket

bind() Bind IP and port

listen() Wait for client

accept() Accept connection

connect() Connect to server

send() Send data

recv() Receive data

close() Close socket

6. Differentiate between TCP Socket and UDP Socket

TCP Socket	UDP Socket
Connection-oriented	Connectionless
Reliable	Unreliable
Ordered delivery	No ordering
Slower	Faster

TCP used in:

- File transfer

UDP used in:

- Video streaming
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7. How communication between client and server done in TCP socket programming**Answer:**

Steps in TCP communication:

Server Side

1. socket()
2. bind()
3. listen()
4. accept()
5. send()/recv()

Client Side

1. socket()
2. connect()
3. send()/recv()

TCP uses:

- 3-way handshake
- Reliable communication

8. List features of Packet Tracer

Answer:

- Network simulation
- Router and switch configuration
- Supports routing protocols
- Easy troubleshooting
- Wireless configuration
- Packet analysis

9. How to design network in Packet Tracer

Answer:

1. Drag devices like PC, router, switch
2. Connect using cables
3. Configure IP addresses
4. Configure routing if needed
5. Test using ping command

10. Explain different types of cables supported in Packet Tracer software

Cable	Use
Straight-through	Different devices
Cross-over	Same devices
Console cable	PC to router configuration
Fiber cable	High-speed communication

11. Write commands for router configuration

enable

configure terminal

```
hostname R1
```

```
interface fa0/0
```

```
ip address 192.168.1.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

```
For RIP:
```

```
router rip
```

```
version 2
```

```
network 192.168.1.0
```

12. What is the Default Gateway?

Answer:

Default gateway is the IP address of router through which devices communicate outside their local network.

Example:

192.168.1.1

13. Differentiate between Router and Switches

Router	Switch
Layer 3 device	Layer 2 device
Uses IP address	Uses MAC address
Connects networks	Connects devices

14. What is DHCP? How to configure it?

Answer:

DHCP automatically assigns IP addresses to devices.

Basic configuration:

```
ip dhcp pool LAN
```

```
network 192.168.1.0 255.255.255.0
```

```
default-router 192.168.1.1
```

```
dns-server 8.8.8.8
```

15. Differentiate between RIP, OSPF and EIGRP

RIP	OSPF	EIGRP
Distance vector	Link state	Hybrid
Hop count	Cost	Bandwidth + Delay
Slow	Fast	Very fast

16. How dynamic routing is performed in EIGRP

Answer:

EIGRP dynamically exchanges routing information between routers using hello packets and DUAL algorithm.

It calculates best route using:

- Bandwidth
- Delay

17. How to configure wireless router

Answer:

1. Open wireless router
2. Set SSID
3. Configure IP
4. Enable DHCP
5. Set WPA2 password
6. Connect wireless devices

18. Write Network Administration Commands

Command Purpose

ping	Check connectivity
ipconfig	View IP config
ifconfig	Linux IP config
tracert	Trace route
netstat	Network statistics
nslookup	DNS lookup

19. How to configure an FTP server?

Answer:

1. Install FTP software like vsftpd
2. Configure FTP settings
3. Create users
4. Start FTP service
5. Connect using FTP client

Default Port:

- 21

20. Explain RSA Algorithm in detail

Answer:

RSA is an asymmetric encryption algorithm.

Steps:

1. Choose two prime numbers p and q

2. Compute:

$$n = p \times q$$

3. Compute:

$$\phi(n) = (p-1)(q-1)$$

4. Choose e such that:

$$\gcd(e, \phi(n)) = 1$$

5. Compute d :

$$d \times e \bmod \phi(n) = 1$$

Public key:

(e, n)

Private key:

(d, n)

Encryption:

$$C = M^e \bmod n$$

Decryption:

$$M = C^d \bmod n$$

21. Explain Diffie Hellman Algorithm in detail

Answer:

Used for secure key exchange.

Steps:

1. Choose prime number p and primitive root g

2. Alice selects private key a

3. Bob selects private key b

Alice computes:

$$A = g^a \bmod p$$

Bob computes:

$$B = g^b \bmod p$$

Exchange A and B .

Shared secret:

$$K = B^a \bmod p$$

or

$$K = A^b \bmod p$$

Both get same key.

22. What do you mean by Primitive Root?

Answer:

Primitive root is a number whose powers generate all numbers from 1 to $p-1$ modulo p .

Used in:

- Diffie Hellman
-

23. What are Coprime Numbers?

Answer:

Two numbers whose GCD is 1.

Example:

8 and 15

24. How are keys generated in RSA?

Answer:

Keys generated using:

1. Prime numbers
2. Euler totient function
3. Public exponent e
4. Private exponent d

Public key:

(e, n)

Private key:

(d, n)

25. Explain logic of Digital Signature RSA cryptosystem implementation

Answer:

1. Sender creates hash of message
2. Encrypts hash using private key
3. Receiver decrypts using sender public key
4. Receiver compares hash

If hashes match:

- Integrity verified
 - Sender authenticated
-

26. What do you mean by Digital Signature?

Answer:

Digital signature is a cryptographic technique used for:

- Authentication
 - Integrity
 - Non-repudiation
-

27. What is Snort?

Answer:

Snort is an open-source Intrusion Detection System (IDS).

Used for:

- Monitoring network traffic
- Detecting attacks

28. Where is the snort.conf file located?

Answer:

Usually located at:

/etc/snort/snort.conf

29. Explain different cyber security attacks

Answer:

Attack	Meaning
Phishing	Fake websites/emails
Malware	Malicious software
SQL Injection	Malicious SQL queries
DDoS	Flooding attack
MITM	Intercepts communication

30. What is a DDoS attack?

Answer:

Distributed Denial of Service attack floods server with huge traffic from multiple systems causing service failure.

31. What is Cyber Security?

Answer:

Cyber security is protection of:

- Networks
- Systems
- Data

from cyber attacks.

32. Differentiate between Stream Cipher and Block Cipher

Stream Cipher	Block Cipher
Encrypts bit-by-bit	Encrypts block-by-block
Faster	More secure
Example: RC4	Example: DES, AES

33. Differentiate between Symmetric and Asymmetric Cipher

Symmetric Asymmetric

Same key Two keys

Faster Slower

DES, AES RSA

34. Explain the working of DES

Answer:

DES is symmetric block cipher.

Features:

- 64-bit block size
- 56-bit key
- 16 rounds

Steps:

1. Initial permutation
 2. 16 Feistel rounds
 3. Final permutation
-

35. Explain steps of RSA algorithm

Answer:

1. Select primes p and q
 2. Compute n
 3. Compute $\phi(n)$
 4. Select e
 5. Compute d
 6. Generate keys
 7. Encrypt/decrypt
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36. Draw block diagram of DES

Answer to tell in viva:

“Sir, DES contains:

- Initial permutation
- 16 Feistel rounds
- Final permutation
- Each round contains expansion permutation, XOR with key, S-box substitution and permutation.”

(Usually enough in viva.)

37. Numerical on RSA and Diffie Hellman

For numericals remember formulas:

RSA:

$$C = M^e \bmod n$$

$$M = C^d \bmod n$$

Diffie Hellman:

$$K = g^{ab} \bmod p$$

38. Linux Network Commands

Command Purpose

ifconfig	Show IP
ping	Test connectivity
netstat	Network stats
traceroute	Trace route
nslookup	DNS lookup

39. Class A, B, C, D, E IP addresses

Class Range

A	1–126
B	128–191
C	192–223
D	224–239
E	240–255

40. Comment on Private and Public IPs

Private IP	Public IP
Internal use	Internet use
Not routable on internet	Globally routable
Private ranges:	
10.0.0.0	
172.16.0.0 – 172.31.0.0	
192.168.0.0	

41. How to configure Access Control List and MAC filtering

ACL Example

access-list 1 permit 192.168.1.0 0.0.0.255

MAC Filtering

Add allowed MAC addresses in wireless router settings.

42. What do you mean by NAT? Use of NAT

Answer:

NAT converts private IP addresses into public IP addresses.

Uses:

- Conserves public IPs
- Security
- Internet access

43. What do you mean by Static NAT and Dynamic NAT

Static NAT	Dynamic NAT
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Fixed mapping	Uses pool
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One-to-one	Temporary mapping
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44. Command to set Static NAT

ip nat inside source static 192.168.1.2 200.1.1.1

45. What do you mean by Multiplicative Inverse?**Answer:**

A number which when multiplied with another number gives 1 modulo n.

Example:

$$3 \times 7 \bmod 10 = 1$$

So:

- 7 is multiplicative inverse of 3 modulo 10.

Used in:

- RSA for finding d value.